

Frozen MDLM: near-absorbing token diagnostic

CPU diagnostic only. Zero training updates, generated molecules, property-oracle calls or GPUs. This is not a molecular quality result or evidence of GenMol superiority.

Same local MDLM50k EMA; 16 reused validation rows, 814 content tokens, padded shape [16,106]. All four MASK mixture weights and three noise times were declared before execution. Corruption seeds 2400/2401/2402 reset across priors at each time; twelve frozen forwards. Four intra-op / one inter-op CPU threads. Producer runtime 28.270s; complete child wall time 41.096s.

MASK weight	Time	All CE	Top-1 %	MASK n / CE	Changed other n / CE	Unchanged n / CE
0.9	0.1	0.356	90.42	69 / 0.289	6 / 0.147	739 / 0.364
0.9	0.5	1.389	71.87	377 / 1.993	42 / 3.021	395 / 0.639
0.9	0.9	3.670	25.80	653 / 3.831	72 / 5.538	89 / 0.979
0.99	0.1	0.346	90.54	77 / 0.283	0 / NA	737 / 0.353
0.99	0.5	1.153	76.04	418 / 1.748	3 / 4.756	393 / 0.494
0.99	0.9	2.857	31.82	719 / 3.100	10 / 4.180	85 / 0.647
0.999	0.1	0.346	90.54	77 / 0.283	0 / NA	737 / 0.353
0.999	0.5	1.047	77.52	423 / 1.576	0 / NA	391 / 0.474
0.999	0.9	2.545	31.94	729 / 2.772	1 / 2.305	84 / 0.583
0.9999	0.1	0.346	90.54	77 / 0.283	0 / NA	737 / 0.353
0.9999	0.5	1.047	77.52	423 / 1.576	0 / NA	391 / 0.474
0.9999	0.9	2.536	32.80	730 / 2.761	0 / NA	84 / 0.585

CE is mean negative log probability of the original clean token (nats); top-1 is token accuracy. At t=0.9, all-content CE changes from 3.670 at weight0.9 to 2.536 at0.9999. Changed non-MASK observations change from 72 to 0. The corruption task changes, and empty/small groups cannot establish reliable conditional performance. All48 group records, CE sums and unreduced logit-gradient norms remain in groups.csv and the raw JSON.

Analytic comparison and interpretation

Let π_E be the smoothed empirical prior, m the MASK ID, λ its mixture weight, and δ_j the point mass at clean token j . $\pi_\lambda = \lambda \delta_m + (1-\lambda) \pi_E$; $q_t(j) = \alpha_t \delta_j + (1-\alpha_t) \pi_\lambda$, with $\alpha_t = 1 - 0.999^t$. The one-token total variation distance to the absorbing marginal is exactly $(1-\alpha_t)(1-\lambda)(1-\pi_E[m])$, regardless of j . All tested priors have positive support.

MASK weight	Mean terminal KL (nats)
0.9	0.000117005
0.99	0.000628705
0.999	0.002018659
0.9999	0.004072913

Terminal KL averages $KL(q_1(j) || \pi_\lambda)$ over the 814 observed clean tokens, with $\alpha_1 = 0.001$. It increases as the prior approaches MASK and diverges at the absorbing limit for fixed positive α_1 . Closer forward marginals alone do not establish equivalent learned reverse trajectories. The model was trained with MDLM masked-target loss; its unrestricted logits at unmasked observations need not be a calibrated clean posterior for the new categorical corruption. No CE-to-LOO conversion or reverse generation was performed here.

This extends an earlier diagnostic on the same reused rows. Common RNG streams are not maximal or nested coupling. One corruption draw per condition supplies no confidence interval. Mixed absorbing transitions are established in D3PM Appendix A.2.6 (<https://arxiv.org/html/2107.03006>). Any zero-update transfer or subsequent training needs a separate checkpoint identity and prospective molecular benchmark.

Provenance

Producer source: e108f1184c6e7fd062e952a1f20f342eb62344dc
MDLM checkpoint SHA-256: 8d00aa47b02f64bf39ff6b0b2e786f213587366fc2c3d29712a00f3f84108dd6
Validation panel SHA-256: e2493da4f3cb3217b48c78dc2901dc7524d7d90dc3959cfa87a1b6f8a9a7658
Token frequency SHA-256: 088c78e75611f3cc42c4011e1da6f65a377e673b9cba07a28b126b0fc62f06ed